



Commands + Code + Text ▶ Run all

RAM Disk



Double-click (or enter) to edit

```
[2] ✓ 5s
!pip install transformers datasets torch accelerate

Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: datasets in /usr/local/lib/python3.12/dist-packages (4.0.0)
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```

Double-click (or enter) to edit

```
[3] ✓ 0s
text = """
Artificial Intelligence is transforming industries worldwide.
Machine learning enables computers to learn from data.
Deep learning powers modern applications such as image recognition.
AI is used in healthcare, finance, education, and transportation.
Generative AI can create text, images, music, and videos.
"""

with open("data.txt", "w") as f:
    f.write(text)

    print(f"Dataset created successfully")

... Dataset created successfully
```



[7]

✓ 0s



```

text = """
Artificial Intelligence is transforming industries worldwide.
Machine learning enables computers to learn from data.
Deep learning powers modern applications such as image recognition.
AI is used in healthcare, finance, education, and transportation.
Generative AI can create text, images, music, and videos.
"""

with open("data.txt", "w") as f:
    f.write(text)

    print("Dataset created successfully")

```



... Dataset created successfully

[8]

✓ 40s



```

from transformers import GPT2Tokenizer, GPT2LMHeadModel

tokenizer = GPT2Tokenizer.from_pretrained("gpt2")
model = GPT2LMHeadModel.from_pretrained("gpt2")

print("GPT-2 loaded successfully")

```



```

... /usr/local/lib/python3.12/dist-packages/huggingface_hub
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional.
warnings.warn(
Warning: You are sending unauthenticated requests to the Hugging Face Hub.
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the Hugging Face Hub.
tokenizer_config.json: 100% ██████████ 26.0/26.0 [00:00<00:00, 811B/s]
vocab.json: 100% ██████████ 1.04M/1.04M [00:00<00:00, 13.1MB/s]
merges.txt: 100% ██████████ 456k/456k [00:00<00:00, 6.92MB/s]
tokenizer.json: 100% ██████████ 1.36M/1.36M [00:00<00:00, 17.3MB/s]
config.json: 100% ██████████ 665/665 [00:00<00:00, 24.0kB/s]
model.safetensors: 100% ██████████ 548M/548M [00:03<00:00, 193MB/s]
Loading weights: 100% ██████████ 148/148 [00:00<00:00, 524.10it/s, Materializing param=transformer.wte.weight]
GPT2LMHeadModel LOAD REPORT from: gpt2
Key | Status |
-----+-----+
h.{0...11}.attn.bias | UNEXPECTED |

Notes:
- UNEXPECTED :can be ignored when loading from different checkpoints.
generation_config.json: 100% ██████████ 124/124 [00:00<00:00, 6.08kB/s]

GPT-2 loaded successfully

```



Commands + Code + Text ▶ Run all ☁

✓ RAM

Disk



Double-click (or enter) to edit

```
[14] ✓ 0s
text = """
Artificial Intelligence is transforming industries worldwide.
Machine learning enables computers to learn from data.
Deep learning powers modern applications such as image recognition.
AI is used in healthcare, finance, education, and transportation.
Generative AI can create text, images, music, and videos.
"""
```

```
with open("data.txt", "w") as f:
    f.write(text)

print("Dataset created successfully")
```

Dataset created successfully

```
[15] ✓ 0s

from datasets import Dataset

with open("data.txt", "r") as f:
    text = f.read()

dataset = Dataset.from_dict({"text": [text]})
dataset
```

```
[16] ✓ 1s

from transformers import GPT2Tokenizer, GPT2LMHeadModel

tokenizer = GPT2Tokenizer.from_pretrained("gpt2")
model = GPT2LMHeadModel.from_pretrained("gpt2")

print("GPT-2 loaded successfully")
```

Loading weights: 100%  148/148 [00:00<00:00, 212.95it/s, Materializing param=transformer.wte.weight]

GPT2LMHeadModel LOAD REPORT from: gpt2

Key	Status	
h.{0...11}.attn.bias	UNEXPECTED	

Notes:

- UNEXPECTED :can be ignored when loading from different task/

GPT-2 loaded successfully

```
[18] ✓ 0s

tokenizer.pad_token = tokenizer.eos_token
```

```
[23] ✓ 0s

def tokenize_function(examples):
    return tokenizer(
        examples["text"],
        truncation=True,
        padding="max_length",
        max_length=128
    )

tokenized_dataset = dataset.map(tokenize_function)

print(tokenized_dataset)
```

Map: 100%  1/1 [00:00<00:00, 16.26 examples/s]
Dataset({
 features: ['text', 'input_ids', 'attention_mask'],
 num_rows: 1
})

```
[24] ✓ 0s

tokenizer.pad_token = tokenizer.eos_token
model.resize_token_embeddings(len(tokenizer))
```

Embedding(50257, 768)

```
[26] ✓ 0s

from transformers import Trainer, TrainingArguments

training_args = TrainingArguments(
    output_dir="./results",
    num_train_epochs=5,
    per_device_train_batch_size=1,
    save_steps=100,
    save_total_limit=1
)
```





```
tokenizer.pad_token = tokenizer.eos_token
```

```
def tokenize_function(examples):
    tokenized_inputs = tokenizer(
        examples["text"],
        truncation=True,
        padding="max_length",
        max_length=128
    )
    tokenized_inputs["labels"] = tokenized_inputs["input_ids"].copy()
    return tokenized_inputs
```

```
tokenized_dataset = dataset.map(tokenize_function, batched=True)
```

```
print(tokenized_dataset)
```

Map: 100%  1/1 [00:00<00:00, 12.47 examples/s]

```
Dataset({
  features: ['text', 'input_ids', 'attention_mask', 'labels'],
  num_rows: 1
})
```

```
tokenizer.pad_token = tokenizer.eos_token
model.resize_token_embeddings(len(tokenizer))
```

Embedding(50257, 768)

```
from transformers import Trainer, TrainingArguments, DataCollatorForLanguageModeling
```

```
data_collator = DataCollatorForLanguageModeling(tokenizer=tokenizer, mlm=False)
```

```
training_args = TrainingArguments(
    output_dir="./results",
    num_train_epochs=5,
    per_device_train_batch_size=1,
    save_steps=100,
    save_total_limit=1
)
```

```
trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=tokenized_dataset,
    data_collator=data_collator
)
```

```
# Temporarily inspect a batch from the data collator
# from torch.utils.data import DataLoader
# dataloader = DataLoader(tokenized_dataset, batch_size=1, collate_fn=data_collator)
# batch = next(iter(dataloader))
# print(batch.keys())
# print({k: v.shape for k, v in batch.items()})
```

```
trainer.train()
```

/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin_memory' argument is super().__init__(loader)

[5/5 00:42, Epoch 5/5]

Step Training Loss

Writing model shards: 100%  1/1 [00:01<00:00, 1.94s/it]

```
TrainOutput(global_step=5, training_loss=2.6250194549560546,
metrics={'train_runtime': 63.8966,
'train_samples_per_second': 0.078, 'train_steps_per_second':
0.078, 'total_flos': 326615040000.0, 'train_loss':
2.6250194549560546, 'epoch': 5.0})
```

```
prompt = "Artificial Intelligence"
```

```
inputs = tokenizer(prompt, return_tensors="pt")
```

```
outputs = model.generate( **inputs, max_length=150, do_sample=True, temperature=0.8, top_k=50, top_p=0.95)
```

```
generated_text = tokenizer.decode(outputs[0], skip_special_tokens=True)
```

```
print(generated_text)
```

```
print(tokenized_dataset.features)
print(tokenized_dataset[0])
```

```
{'text': Value('string'), 'input_ids': List(Value('int32')), 'attention_mask': List(Value('int8')), 'labels': List
{'text': '\nArtificial Intelligence is transforming industries worldwide.\nMachine learning enables computers to l
```